

Narrative Quantification: Compiling Narrative Structures into Geometric State Representations

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Abstract

Human societies interpret events through structured narratives rather than isolated data points. This paper extends the framework introduced in Nakamura (2026, SSRN 6419019). We formalize Narrative Quantification as a computational framework for representing, transforming, and verifying such interpretations as structured states.

The framework decomposes narratives into structured components—actors, events, causal relationships, conflicts, and outcomes—which are encoded into a high-dimensional structural representation (24TAG). These representations are deterministically transformed into a geometric state space via a mapping Φ , enabling narratives to be analyzed within a measurable metric space. A cryptographic state hash ensures invariance and verifiability of narrative states under representational transformations.

Building on this foundation, we introduce the Narrative Compiler, which converts unstructured text into structured cognitive states, and define Large Cognitive Models (LCMs) that perform reasoning as transformations over probability distributions of these states, rather than over token sequences.

We argue that Narrative Quantification establishes a new cognitive infrastructure, enabling AI systems to operate not only on language, but on the underlying structure of interpretation itself.

1. Introduction

Most contemporary AI systems rely on probabilistic language modeling. Large Language Models (LLMs) generate text by predicting token probabilities within massive corpora. While effective for language generation, these systems do not explicitly represent or operate on structured narrative states, causal relationships, or long-term narrative dynamics.

Human cognition, however, rarely processes information as isolated tokens. Instead, humans organize knowledge through **narratives**: structured relationships between actors, events, motivations, and outcomes.

We propose and formalize **Narrative Quantification** as a computational framework for representing, transforming, and verifying narrative interpretations as structured states. By decomposing narratives into structured components—such as actors, events, causal relationships, conflicts, and outcomes—it becomes possible to construct databases of collective narrative patterns. This enables

narrative interpretations to be treated as measurable, comparable, and computationally tractable objects.



Figure 1: Figure 1: Processing pipeline comparison between LLM (token-based) and LCM (narrative-state-based).

2. Narrative Cognition

Human cognition frequently follows a narrative processing loop:

$$\text{Event} \rightarrow \text{Meaning} \rightarrow \text{Narrative} \rightarrow \text{Memory}$$

Experiences are rarely stored as raw data. Instead, individuals interpret events through causal

explanations and emotional context, forming narratives that are later stored in memory.

This loop can be formalized as a transformation pipeline over structured representations. This process can be interpreted as a sequence of transformations over structured states, where complex experiences are compressed into lower-dimensional representations. This mechanism allows humans to encode information in a form that supports prediction, generalization, and decision-making.

Narrative cognition therefore represents a fundamental cognitive process rather than a mere cultural artifact. This motivates the need for a formal representation of narrative structure, which we introduce in the following sections.

3. Narrative Quantification

Narrative Quantification is the formal process of converting unstructured narrative text into analyzable, high-density cognitive data. This transformation is achieved through a multi-stage pipeline: **Decomposition**, **Event Formation**, and **Narrative Compression**.

3.1 Decomposition

The first stage segments raw text into primary semantic units. Rather than treating a story as a continuous stream of words, we identify discrete entities and their roles. By isolating **Actors** (agents of change), **Events** (actions), and **Causal Relationships** (the ‘why’), we transform linguistic ambiguity into a set of distinct variables.

3.2 Event Formation: The Minimum Causal Interaction Unit

Continuous occurrences are discretized into logical structures. Within this framework, an event is defined as the **minimum causal interaction unit** between actors unfolding over time. Unlike traditional data processing, event formation recognizes the inherent unity of action and reaction. A typical atomic event unit includes:

- **Actors:** The agents initiating or receiving the interaction.
- **Action-Reaction Pair:** The core causal loop (e.g., Actor A exerts influence, Actor B retaliates or transitions).
- **Temporal Boundary:** The window from the initial trigger to the resulting outcome.

This shift from **tokens to events** enables Large Cognitive Models (LCM) to reason over the logic of interaction itself.

3.3 Narrative Compression

Compression is an active **entropy reduction** process. Raw data is inherently high-entropy (noisy and continuous). By filtering for the structural core, we achieve:

- **Computational Efficiency:** Reasoning over compressed event units is faster than processing raw tokens.
- **Long-Term Coherence:** Stripping noise makes the “narrative arc” visible across long temporal scales.

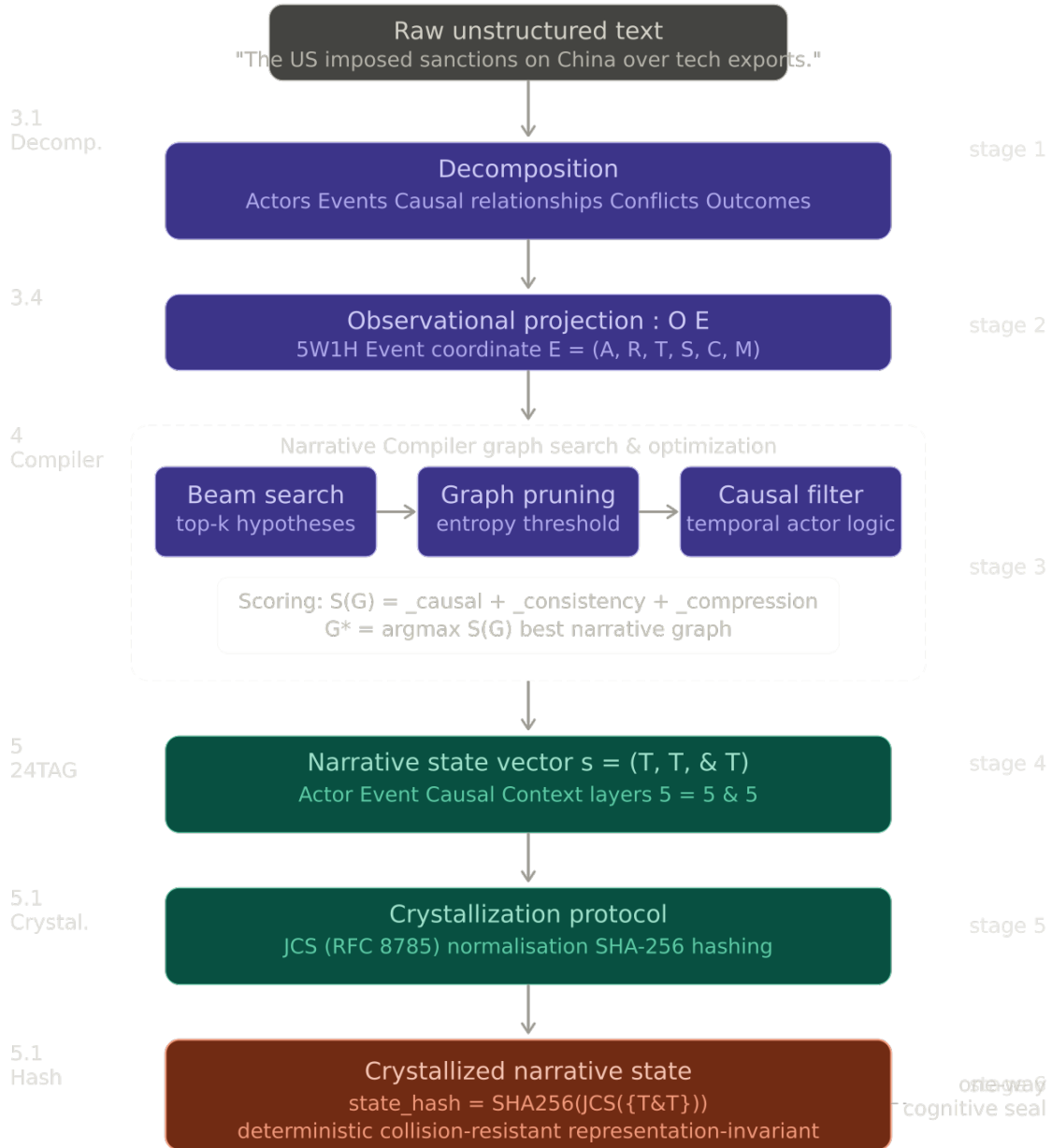


Figure 2: Figure 2: Narrative compilation pipeline from raw text to crystallized state hash.

- **Knowledge Crystallization:** Transforming “text” into “structured knowledge” for the LCM.
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3.4 Observational Projection: 5W1H to Event Coordinates

To bridge the gap between descriptive journalism (5W1H) and computational reasoning, we introduce **Observational Projection**.

3.4.1 The First Mapping: Φ_1 We define an observation extracted from text as:

$$O = (\text{who, what, when, where, why, how})$$

We define the projection as Φ_1 to distinguish it from the subsequent geometric embedding Φ_2 introduced later in the pipeline:

$$\Phi_1 : O \rightarrow E$$

where E represents the structured **Event Coordinate**.

3.4.2 Event Coordinate Structure The event coordinate E is a six-dimensional tuple:

$$E = (A, R, T, S, C, M)$$

- **A (Actor Set):** Participating agents $A = \{a_1, a_2, \dots, a_n\}$.
- **R (Relationships):** Directed interaction edges $(a_i \xrightarrow{\tau} a_j)$ with relation types τ (e.g., *cooperate*, *sanction*).
- **T (Temporal Vector):** $T = (t_{\text{cause}}, t_{\text{event}}, t_{\text{impact}})$, distinguishing the origin, interaction, and consequence.
- **S (Spatial Vector):** $S = (s_{\text{origin}}, s_{\text{event}}, s_{\text{impact}})$, tracking the propagation across physical and decision domains.
- **C (Causal Classification):** Derived from the temporal ordering and interaction pattern.
- **M (Mechanism Category):** The specific modality of the interaction.

3.4.3 Implementation as a Probabilistic Estimator In practice, Φ_1 is implemented as a probabilistic estimator:

$$\hat{E} = \Phi_1(O; \theta)$$

This hybrid model combines large language model inference with rule-based constraints to identify candidates, classify semantics, and enforce causal plausibility (e.g., temporal precedence).

3.4.4 Integration into the Pipeline The resulting event coordinates are subsequently encoded into the **24TAG structural representation**, forming the basis for narrative state construction and the generation of the **Narrative Graph** described in Section 4.

4. The Narrative Compiler: Search and Optimization

The **Narrative Compiler** is the central engine of the framework. It does not merely extract data; it explores the vast space of possible narrative graphs $\mathcal{G}$ to identify the most probable narrative structures $G \in \mathcal{G}$. Given that a single text can yield an exponential number of event connections, the Compiler treats narrative construction as a **graph-search problem**.

4.1 Generating Possible Narrative Graphs

From a set of extracted events E , the Compiler builds a **Narrative Graph (G)**, where nodes represent events and edges represent causal or temporal transitions. To manage complexity and prevent the “hallucination” of illogical connections, the Compiler employs specific Search Constraints to generate the final probability distribution $P(S \mid T)$.

4.2 Beam Search for Hypothesis Management

To maintain a manageable “hash set” of narrative states, the Compiler utilizes **Beam Search**. Rather than exploring every possible graph configuration, the Compiler keeps only the top- k most plausible narrative hypotheses at each step of the construction.

- **Top- k Survival:** At each stage of event linking, only the graphs with the highest cumulative probability scores are retained.
- **Diversity Maintenance:** The search preserves divergent narrative paths if their probabilities are sufficiently high, ensuring that the final distribution captures legitimate ambiguity.

4.3 Graph Pruning and Information Entropy

Graph Pruning is the active removal of low-probability or redundant branches in the narrative tree.

- **Redundancy Filtering:** If multiple paths lead to the same structural state with lower efficiency, they are pruned to reduce computational overhead.
- **Entropy Thresholding:** Branches that increase the conditional entropy of the narrative without adding significant semantic value are eliminated, facilitating **Narrative Compression**.

4.4 The Causal Filter: Logical Constraints

The most critical optimization is the **Causal Filter**, which acts as a structural constraint system. The Compiler applies a set of logical rules to validate every proposed edge:

1. **Temporal Precedence:** A cause must strictly precede its effect in the temporal dimension.
2. **Actor Consistency:** Actions must align with the capabilities and motivations of the involved actors.

3. **Interaction Logic:** The Action–Reaction loop must maintain a coherent flow.

4.4.1 Narrative Graph Scoring Function The constraints are integrated into a unified scoring function $S(G)$ that evaluates each candidate graph for plausibility and structural efficiency:

$$S(G) = \alpha\Gamma_{\text{causal}} + \beta\Lambda_{\text{consistency}} + \gamma\Omega_{\text{compression}}$$

This scoring function induces a probability distribution over the set of possible graphs:

$$P(G \mid T) \propto \exp(S(G))$$

This formulation connects graph optimization directly with probabilistic inference.

4.5 Final State Synthesis and Crystallization

After iterative graph expansion, the Compiler selects the highest-scoring narrative graph G^* from the beam search candidate pool $\mathcal{B}$:

$$G^* = \arg \max_{G \in \mathcal{B}} S(G)$$

The selected graph is transformed into a **Narrative State Vector** $\mathbf{s}$ by aggregating the attributes of its constituent event units according to the **24TAG** schema:

$$\mathbf{s} = (T_1, T_2, \dots, T_{24})$$

The resulting state vector $\mathbf{s}$ defines a point in the narrative state space $\mathcal{S}$, enabling downstream probabilistic reasoning in the **LCM framework**. To ensure reproducibility, the state vector is deterministically serialized (e.g., via JCS) and converted into a cryptographic fingerprint:

$$h = H(\text{serialize}(\mathbf{s}))$$

where H is a cryptographic hash (e.g., SHA-256). This **state hash** h uniquely identifies the semantic interpretation, allowing independent systems to verify and compare narrative analyses.

5. The 24TAG Taxonomy

To ensure universal compatibility and deterministic hashing, the Narrative Quantification protocol defines a fixed schema of **24 structured tags**. These tags are categorized into four primary layers, representing the essential dimensions of human narrative cognition.

Table 1: The 24TAG Framework Definitions

Layer	Tag ID	Tag Name	Description
Actor & Agency	T01	Primary_Actor	The main entity initiating the interaction.
	T02	Secondary_Actor	The entity receiving or responding to the action.
	T03	Actor_Role_A	Functional role of the primary actor (e.g., Protagonist).
	T04	Actor_Role_B	Functional role of the secondary actor (e.g., Antagonist).
	T05	Actor_Motivation_A	The underlying “Why” for Actor A.
	T06	Actor_Motivation_B	The underlying “Why” for Actor B.
Event & Action	T07	Action_Type	Category of the event (e.g., Verbal, Physical, Financial).
	T08	Action_Intensity	Magnitude of the action (Scaled 0.0 to 1.0).
	T09	Target_Resource	The specific object or value being contested.
	T10	Event_Modality	Whether the event is factual, hypothetical, or desired.
	T11	Temporal_Sequence	Relative order of the event within the narrative arc.
	T12	Spatial_Context	Domain or environment where the event occurs.
Causal & Logic	T13	Causal_Link_Type	Nature of the connection (e.g., Direct Cause, Enabling).
	T14	Conflict_Nature	Type of opposition (e.g., Internal, Interpersonal).
	T15	Conflict_Intensity	Severity of the friction between actors.
	T16	Resolution_Status	Degree to which the conflict has been settled.

Layer	Tag ID	Tag Name	Description
Context & Tone	T17	Outcome_Valence	Positive or negative impact on the narrative state.
	T18	Logic_Consistency	Degree of alignment with prior narrative states.
	T19	Emotional_Tone_A	Emotional state of Actor A during the event.
	T20	Emotional_Tone_B	Emotional state of Actor B during the event.
	T21	Perspective_Bias	Point of view from which the narrative is told.
	T22	Info_Asymmetry	Gap in knowledge between the actors involved.
	T23	Significance_Score	Importance of this event to the overall arc.
	T24	Narrative_Closure	Extent to which this event concludes a specific thread.

5.0 Rationale for the 24-Tag Schema

The choice of exactly 24 structural tags is driven by two primary considerations:

1. **Cognitive Inspiration:** The set aligns with the approximate complexity threshold that human cognition can integrate when processing causal narratives.
2. **Empirical Optimization:** Informed by decomposition experiments, this schema maximizes inter-annotator agreement and provides enough granularity for **Large Cognitive Models (LCM)** while maintaining computational tractability.

5.1 Data Structure and Compilation

We define the 24TAG representation as a structured vector:

$$\mathbf{T} = (T_1, T_2, \dots, T_{24}) \in \mathcal{T}$$

where each component T_i belongs to a predefined domain $\mathcal{D}_i$. The tag space $\mathcal{T}$ is constrained such that each dimension T_i encodes a specific structural property of the narrative, ensuring consistency

and comparability across states. The vector $\mathbf{T}$ uniquely defines a narrative state S , serving as the canonical representation for downstream reasoning.

Each tag T_i is populated by the Narrative Compiler using the **Beam Search** and **Causal Constraints** defined in Section 4. The resulting structure is a JSON object.

The Crystallization Protocol To ensure that a narrative state is immutable and verifiable, we apply a **One-Way Cognitive Seal** via cryptographic hashing.

1. **Normalization (JCS)**: Following RFC 8785 (JSON Canonicalization Scheme), tags are sorted into a canonical format to ensure whitespace or key ordering do not alter the hash.
2. **Self-Referential Exclusion**: The `state_hash` field itself is strictly excluded from the hash input to prevent circular dependencies.
3. **Hashing (SHA-256)**: The Compiler applies the SHA-256 algorithm to the canonical byte-stream.

The mathematical definition of the state hash is:

$$\text{state_hash} = \text{SHA256}(\text{JCS}(\mathbf{T} \setminus \{\text{state_hash}\}))$$

5.2 Integrity and Verification

Because the hash is deterministic, any receiver can verify the integrity of a narrative state by re-calculating the hash from the vector $\mathbf{T}$ and comparing it to the provided `state_hash`. Any divergence discovered during this verification process indicates that the narrative structure has been modified or corrupted post-compilation.

5.3 Formal Type Definitions and Constraints

To bridge the gap between theoretical narrative structure and a machine-executable API, we define the **Type Domains** $\mathcal{D}_i$ for the 24TAG vector $\mathbf{T}$.

The **24TAG Space** is formally defined as the Cartesian product of these individual tag domains:

$$\mathcal{T} = \mathcal{D}_1 \times \mathcal{D}_2 \times \dots \times \mathcal{D}_{24}$$

Table 2: 24TAG Domain and Constraint Specification

Tag ID	Data Type	Domain ($\mathcal{D}_i$)	Description
T01-02	string	UUID / Entity ID	Unique identifier for actors.
T03-04	enum	{Protagonist, Antagonist, Neutral, ...}	Functional narrative roles.
T05-06	string	Natural Language / Embedding	The underlying intent or “Why.”

Tag ID	Data Type	Domain (Di)	Description
T07	enum	{Verbal, Physical, Financial, Cyber, ...}	Primary interaction modality.
T08	float	[0.0, 1.0]	Normalized action intensity.
T10	enum	{Fact, Hypothesis, Desire, Counterfactual}	Epistemic status of the event.
T11	int	$\mathbb{Z}_{\geq 0}$	Sequential index within the graph G .
T13	enum	{Direct, Enabling, Inhibiting, Trigger}	Causal link classification.
T15	float	[0.0, 1.0]	Degree of friction between actors.
T16	int	{0, 1, 2, 3}	Resolution status index.
T17	float	[-1.0, 1.0]	Valence of the impact.
T23	float	[0.0, 1.0]	Event significance weight.
T24	float	[0.0, 1.0]	Narrative closure score.

5.4 Validity and Cross-Tag Constraints

A narrative state $S \in \mathcal{T}$ is considered valid if and only if it satisfies both individual domain constraints and **Cross-Tag Constraints**:

$$S \in \mathcal{T}_{\text{valid}} \subseteq \mathcal{T}$$

Cross-Tag Constraints ensure structural and logical consistency across dimensions. For example:

- **Causal-Temporal Alignment:** T_{13} (Causal Link) must align with T_{11} (Temporal Sequence); a cause cannot follow its effect.
- **Resolution-Valence Coherence:** T_{16} (Resolution Status) must be logically consistent with T_{17} (Outcome Valence) within the context of the specific T_{14} (Conflict Nature).

6. Geometric Definition of Narrative Space

6.1 Narrative Space (Manifold)

Narrative space is defined as a high-dimensional manifold representing the set of all possible narrative states that can be derived from observed events.

Formally, let G denote a narrative graph constructed from a set of events and their causal relations. We define an embedding function

$$e : G \rightarrow \mathbb{R}^d$$

which maps each narrative graph into a point in a d-dimensional vector space.

Under this formulation, each narrative state corresponds to a point on the **narrative manifold**:

$$N = e(G) \in \mathbb{R}^d$$

The collection of all such embedded narrative states forms the **narrative space**.

This geometric representation enables the comparison, clustering, and trajectory analysis of narrative interpretations within a unified mathematical framework.

6.2 Narrative Trajectories

Narratives are not static entities.

As new events occur and interpretations evolve, narrative graphs are updated over time.

This process generates **narrative trajectories** within the narrative manifold:

$$T = e(G_t) \mid t = 1, 2, \dots, T$$

where G_t denotes the narrative graph at time t .

These trajectories capture the temporal evolution of interpretations, enabling the analysis of narrative drift, stabilization, or convergence.

6.3 Quantifying Narrative Divergence

Differences between interpretations can be quantified as distances within the narrative manifold.

One possible distance metric is the Euclidean distance between narrative embeddings:

$$d(G_1, G_2) = \|e(G_1) - e(G_2)\|_2$$

where G_1 and G_2 represent two alternative narrative graphs constructed from the same or related events.

This distance provides a quantitative measure of **narrative divergence**, allowing objective comparison between competing interpretations.

Alternative distance metrics (e.g., cosine similarity or geodesic distances on the manifold) may also be employed depending on the embedding structure.

6.4 Interpretation Geometry

By embedding narrative graphs into a geometric state space $\mathbb{R}^3$ via the mapping Φ , interpretations become mathematically analyzable objects.

The mapping Φ operates exclusively on structural features and is invariant to semantic interpretation.

In this framework, the state $P(x, y, z)$ is defined by structural signal strength, alignment, and intensity, ensuring a non-semantic representation.

The state space is defined as a metric space $(\mathbb{R}^3, \|\cdot\|_2)$, enabling consistent measurement of divergence.

This geometric formulation allows for the following analytical operations:

- **Divergence and Rifts:** Quantifying the Euclidean distance between narrative embeddings: $CDC = \|v_i - \bar{v}\|_2$. This enables detection of Narrative Rifts and measurement of cognitive consensus.
- **Dynamic Phase Transitions (PCE):** Identifying transitions from a collapsed state to an active regime through a discontinuous jump in variance σ^2 . This process exhibits hysteresis, where the reactivation threshold differs from the collapse condition.
- **Structural Trajectory Analysis:** Detecting anomalous narrative paths via trajectory velocity: $v(t) = dP(t)/dt$. This enables trajectory analysis under continuous-time approximation, independent of semantic content.
- **Manifold Convergence:** Analyzing polarization across diverse populations as clustering behavior on a geometric manifold.

Together, these operations establish interpretation as a measurable geometric process.

7. Cognitive Infrastructure

Once narrative data can be systematically compiled and structured, the resulting system evolves beyond simple data storage into what we define as a **cognitive infrastructure**. A cognitive infrastructure is a system that stores and operates on structured representations of interpretation, rather than raw factual data. Traditional information infrastructures focus on storing raw facts. In contrast, the proposed layer manages the contextual and logical structure of information, capturing how events are interpreted within narrative frameworks.

7.1 From Knowledge Bases to Narrative Databases

Conventional knowledge bases excel at storing factual attributes—dates, locations, and entities—representing what happened. However, they generally lack mechanisms for recording how events are interpreted. The Narrative Compiler enables the creation of a **Narrative Database** that stores **Crystallized Narrative States**. A crystallized narrative state is defined as a deterministic mapping from structural features T to a state representation S .

A Crystallized Narrative State preserves:

- **Causal logic:** The underlying relationships and “why” between discrete events.

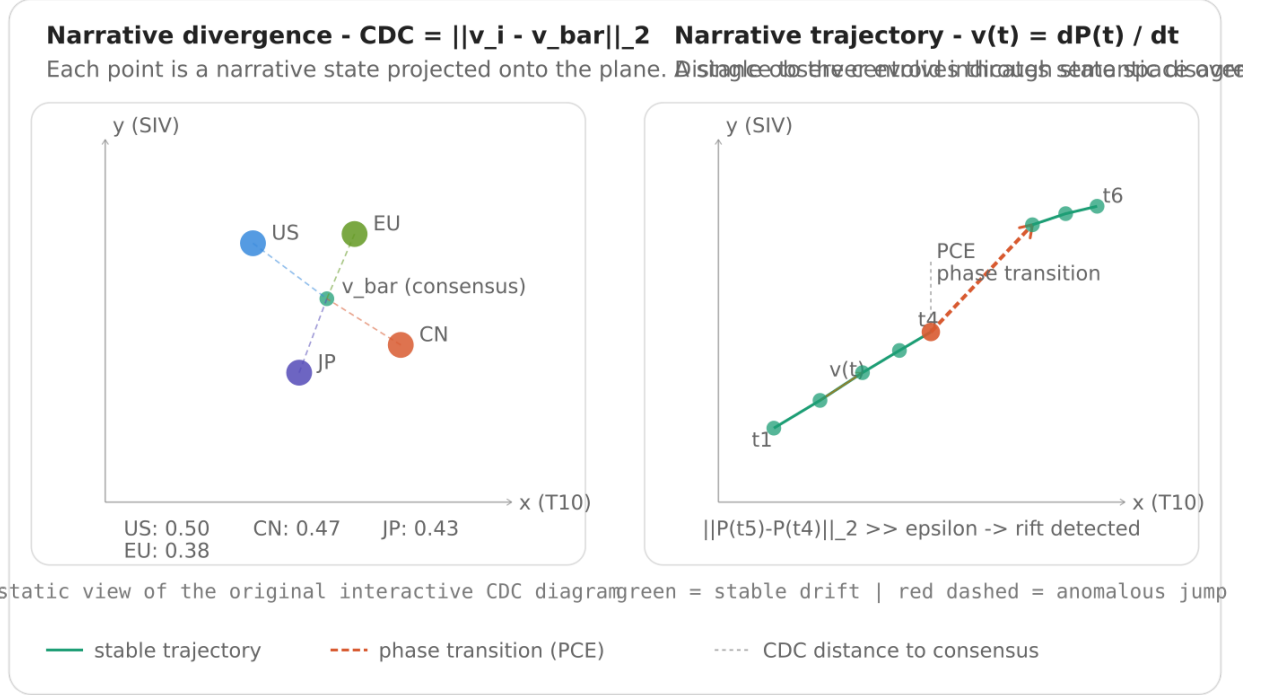


Figure 3: Figure 3: Narrative divergence (CDC) and trajectory in geometric state space.

- **Emotional weighting:** The valence and intensity of interactions between actors.
- **Conflict structures:** The nature and severity of friction between entities.
- **Temporal snapshots:** The state of interpretation captured at a specific point in time.

This represents a fundamental shift from data storage to **cognitive state storage**.

7.2 Information Integrity and the State Hash

The deterministic nature of the **state_hash** provides a mechanism for information integrity within the cognitive infrastructure. By applying the **Crystallization Protocol**—using JSON Canonicalization Scheme (RFC 8785) normalization and SHA-256 hashing—the system creates a **one-way cognitive seal**.

$$state_hash = \text{SHA256}(\text{JCS}(\{T_{01..24}\} \setminus \{state_hash\}))$$

Properties of the State Hash

- **Determinism:** Identical inputs T produce identical hashes.
- **Collision resistance:** Distinct narrative states produce distinct hashes with high probability.
- **Representation invariance:** The hash remains invariant under rendering or presentation transformations.

Interpretation

- **Proof of Interpretation:** A `state_hash` functions as a verifiable fingerprint of a specific structural interpretation.
- **Invariance:** The underlying state remains unchanged regardless of rendering style, preventing manipulation through presentation or framing.

7.3 Applications of the Infrastructure

The proposed cognitive infrastructure supports several applications by leveraging two core properties: (1) **deterministic state representation** (`state_hash`), and (2) **geometric comparability** within the state space $\mathbb{R}^3$. Comparability is defined under a shared metric space $(\mathbb{R}^3, \|\cdot\|_2)$, ensuring consistency across observations.

- **Media Integrity Analysis:** Automated detection of narrative inconsistencies is formulated as a hash divergence problem.

For identical event IDs, differing `state_hash` values indicate structurally distinct interpretations across information sources.

A mismatch in `state_hash` under identical event IDs constitutes a formal inconsistency condition, enabling objective identification of framing or manipulation.

Geopolitical Narrative Mapping : The system monitors how different nations or communities interpret the same international event.

Narrative Rifts are mathematically represented as high-distance clusters in the geometric state space.

Let $D(t) = \|P_i(t) - P_j(t)\|_2$ represent the distance between two interpretations.

Rapid growth in $D(t)$ indicates emerging narrative polarization, allowing early detection of potential conflicts before physical escalation.

Economic and Corporate Intelligence : The infrastructure tracks the evolution of market narratives by observing how structured events and their associated SIV reshape interpretation within the state space.

This enables causal tracing between discrete events and shifts in market interpretation dynamics.

Information Forensics : Reconstructing narrative evolution by tracing state transitions over time enables post-hoc analysis of how and when specific interpretations diverged from the consensus.

Together, these applications demonstrate that narrative interpretation can be detected, compared, predicted, and reconstructed within a unified geometric framework.

8. Reasoning over Narrative Distributions (LCM)

By representing narrative interpretations as probability distributions over narrative states, the proposed **Large Cognitive Model (LCM)** performs reasoning as transformations of distributions. Let $S \in \mathcal{S}$ denote a narrative state within the geometric state space induced by the mapping Φ . A text T induces a probability distribution $P(S | T)$, $S \in \mathcal{S}$, which represents the system’s belief over possible narrative interpretations.

8.1 Inference as Distribution Transformation

The LCM performs inference by transforming distributions over states. Let I_t denote newly observed structured information derived from incoming data. Given a current distribution $P(S_t | T_t)$ and I_t , the system evaluates an updated distribution $P(S_{t+1})$, representing the evolution of narrative interpretations over time.

8.2 Handling Ambiguity and Parallel Narrative Hypotheses

The system maintains a distribution over candidate states $\{(S_i, h_i, P(S_i))\}$, where h_i is the state hash. This enables parallel tracking of narrative hypotheses, allowing the model to manage ambiguity without premature collapse into a single interpretation.

8.3 Reasoning over Causal Trajectories

Inference is modeled as a sequence of state transitions $S_t \rightarrow S_{t+1} = F(S_t, I_t)$, where $F : \mathcal{S} \times I \rightarrow \mathcal{S}$ is the narrative update function. Each transition produces a new state hash $h_{t+1} = H(S_{t+1})$. A structurally inconsistent transition is formally detected when:

$$d(S_{t+1}, \mathbb{E}[S_{t+1}]) > \epsilon$$

where ϵ is a deviation threshold in the metric space.

8.4 Mathematical Framework of LCM Inference

The inference process is formulated as a probabilistic state transition:

$$P(S_{t+1}) = \sum_{S_t} P(S_{t+1} | S_t, I_t) P(S_t)$$

The update is governed by Bayesian inference to ensure the distribution is mathematically closed:

$$P(S_{t+1}) = \frac{1}{Z} P(I_t | S_{t+1}) \sum_{S_t} P(S_{t+1} | S_t) P(S_t)$$

where Z is a normalization constant.

Inference Procedure

1. **Trajectory Prediction:** Predict the future state $\hat{S}_{t+1}$ within the geometric state space.
2. **Consistency Check:** Compute the deviation $d(\hat{S}, S_{obs})$ between predicted and observed states.
3. **Distribution Update:** Update the probability distribution $P(S)$ via Bayesian inference and refine the active candidate set.

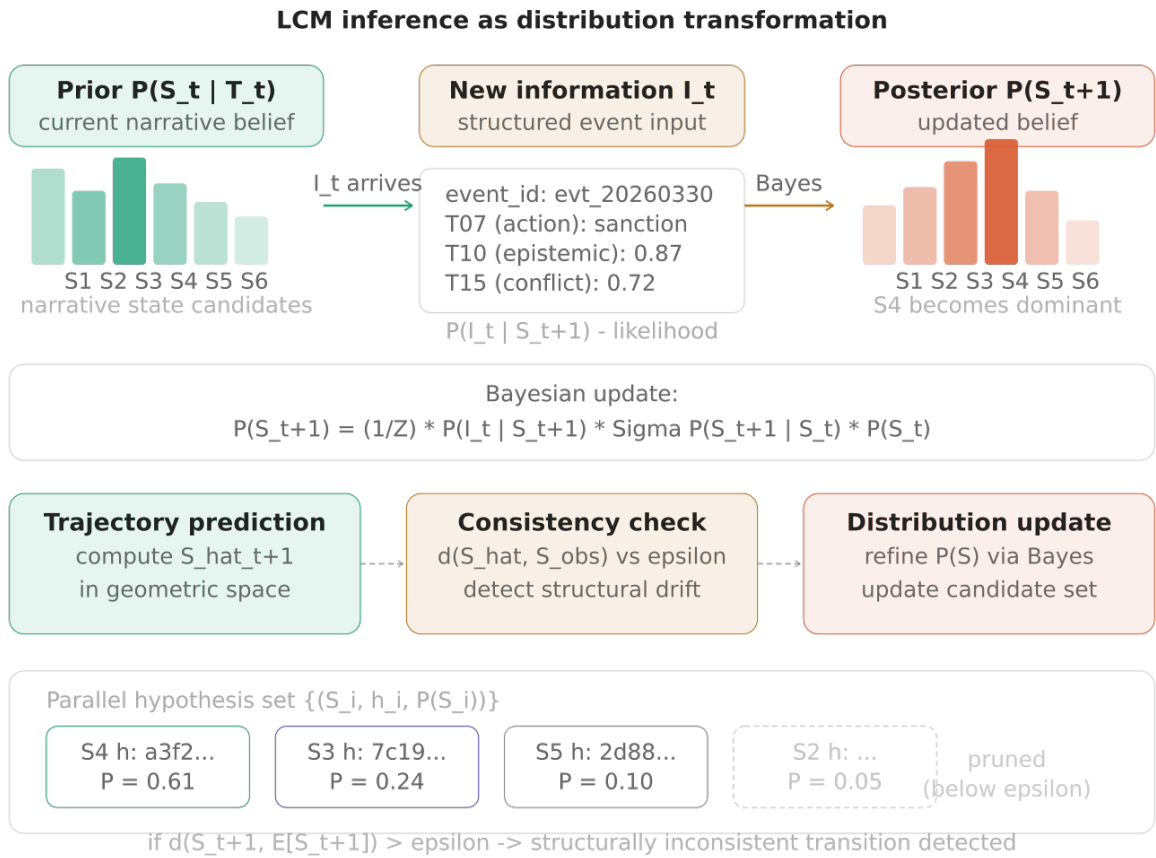


Figure 4: Figure 4: LCM inference as Bayesian distribution transformation over narrative state space.

9. Human Memory and Narrative Cognition

Human memory does not function as a raw sensory log of experience. Instead, we **hypothesize** that memory operates through sophisticated processes of reconstruction and abstraction. Within the Narrative Quantification framework, this phenomenon can be interpreted as a form of **narrative compression**, in which complex experiences are distilled into structured representations.

9.1 Memory as a Compiled Distribution

We hypothesize that human memory can be modeled as maintaining a probability distribution over narrative states $S \in \mathcal{S}$, defined over the geometric state space induced by Φ :

$$P(S \mid \text{Experience})$$

In this model, the brain does not store every perceptual frame; instead, experiences are encoded into structured representations that capture essential relationships between actors, causes, and outcomes. Recall corresponds to reconstructing the most probable narrative structure:

$$S^* = \arg \max_S P(S \mid \text{Experience})$$

This S^* represents the compressed narrative structure that best explains the observed experience.

9.2 Semantic Primitives of Biological Storage

For efficiency, human cognition appears to prioritize a limited set of semantic primitives when encoding experiences:

- **Key Actors:** Identification of the primary agents involved in the interaction.
- **Causal Interpretations:** Assignment of explanations that connect events into coherent cause-effect structures.
- **Emotional Evaluation:** Tagging events with affective valence, influencing prioritization in memory.
- **Outcome:** Encoding the final state or resolution of an interaction.

These primitives closely resemble the structural components used in the **24TAG framework**.

9.3 Biomimetic Interpretation

We hypothesize that the framework is **biomimetic**, mirroring properties associated with human narrative cognition. Representing narrative interpretations through **state hashes** resembles how biological memory systems consolidate complex experiences into stable conceptual representations.

Cognitive Dissonance as Metric Distance

We hypothesize that discrepancies between new experiences and stored narrative structures resemble cognitive dissonance. In the geometric state space, this is formally defined as the Euclidean distance between a stored narrative embedding and a newly observed state:

$$D = \|S_{\text{memory}} - S_{\text{new}}\|_2$$

When $D > \epsilon$, where ϵ is a predefined threshold in the state space, the system attempts to reconcile the discrepancy by updating the internal narrative distribution.

Minimization Principle

This continual updating process can be interpreted as **minimizing expected prediction error** within the narrative state space. This aligns the framework with theories in neuroscience that view cognition as an inference process aimed at minimizing informational entropy.

Closing Line

This suggests that both artificial and biological cognition can be understood as processes of maintaining and updating probability distributions over structured state spaces.

10. Limitations

While the Narrative Quantification framework presents a promising approach to structured cognitive reasoning, several important limitations must be addressed. These limitations arise directly from the structural, geometric, and probabilistic assumptions underlying the framework.

10.1 Subjectivity and Algorithmic Bias

A central challenge in narrative quantification arises from the inherent subjectivity of interpretation. Bias can be interpreted as a distortion in the prior distribution $P(S)$, which is heavily influenced by the training data of the underlying models. This may result in narrative states that are structurally consistent and mathematically valid yet still reflect culturally skewed or biased interpretations.

10.2 Computational Complexity of Graph Generation

Constructing candidate narrative graphs is computationally intensive. For n extracted events, the number of potential causal edges scales as $O(n^2)$, and the total number of possible graph configurations can grow combinatorially. Approximation strategies—such as pruning, sampling, or beam search—are therefore required to maintain computational tractability for real-time applications.

10.3 Cultural and Linguistic Heterogeneity

Narrative structures are not universal. Some cultural traditions emphasize circular or cyclical patterns, while others rely on linear, conflict-driven arcs. A framework optimized for one specific structure may fail to capture the nuances of another, leading to high narrative divergence that reflects cultural differences rather than factual disagreement.

10.4 Data Sovereignty and Strategic Narrative Influence

The construction of a global narrative database raises critical governance questions. If centralized authorities gain the power to define “standard” narrative states for global events, alternative or minority interpretations could be marginalized as anomalous deviations. This raises significant concerns regarding how narrative infrastructures could be utilized for strategic influence or centralized information control.

10.5 Semantic Loss during Compression

Narrative compression inherently involves information reduction. To reach crystallized narrative states, certain elements of the original raw text must be discarded. Subtle aspects of meaning—such as irony, subtext, or poetic ambiguity—remain difficult to preserve when transforming rich prose into structured narrative states. Maintaining a balance between structural clarity and semantic richness is an ongoing challenge.

10.6 Cognitive Local Optima and Confirmation Bias

We hypothesize that the principle of minimizing prediction error can lead to **cognitive local optima**. This corresponds to a mathematical convergence toward a local maximum in $P(S)$, where alternative states and divergent narrative paths are insufficiently explored. This manifests as a mathematical form of confirmation bias, where the system becomes overly resilient to contradictory evidence. Mitigation requires the intentional maintenance of high-entropy divergent hypotheses, even when their current probability is low.

11. Conclusion and Future Work

We formalize **Narrative Quantification** as a computational framework for representing, transforming, and verifying interpretations as structured states. By elevating the fundamental unit of information from tokens to events, and encoding interactions into **crystallized narrative states** via the **24TAG structural hash**, the framework enables the objective verification and prediction of interpretations.

The embedding of narrative graphs into a geometric state space allows interpretations to be analyzed within a measurable metric framework. The framework unifies structure (24TAG), geometry (Φ mapping), and dynamics (distribution and transition) into a single mathematical representation of interpretation.

This shift marks a transition from AI systems that merely generate language to systems that model the structural dynamics of interpretation. By ensuring the **invariance** of narrative states under representational transformations, the framework establishes a deterministic foundation for cognitive reasoning.

Future Work

Future research will focus on two primary trajectories:

- **Bias Mitigation:** Developing “compiler ensembles” that incorporate diverse cultural priors to reduce systemic bias in the state-generation process.
- **Decentralized Governance:** Exploring the decentralized management of narrative hashes to ensure that this infrastructure supports collective interpretation rather than centralized control.

Final Line

Narrative Quantification establishes a foundation for AI systems that operate not on language alone, but on the underlying structure of cognition itself.